Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Andrew Zylstra Dairy 1

Physical address of dairy:

Number and Street	City	County	Zip Code
2655 W Monte Vista AVE	Modesto	Stanislaus	95358

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 12/01/1980

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X022-X002-X016-XXXX X022-X002-X062-XXXX X022-X002-X063-XXXX

B. OPERATORS

Zylstra, Michael John

Operator name: Zylstra, Michael John

Telephone no.: Landline: ; Cellular: (209) 535-2883

Mailing Address Number and Street	City	State	Zip Code
2655 W Monte Vista AVE	Modesto	CA	95358

This operator is responsible for paying permit fees.

C. OWNERS

Zylstra, Michael John

Legal owner name: Zylstra, Michael John

Telephone no.: Landline: ; Cellular: (209) 535-2883

Mailing Address Number and Street	City	State	Zip Code
2655 W Monte Vista AVE	Modesto	CA	95358

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	60	0	0	0	0
Number under roof	1,400	0	0	0	0	0
Maximum number	1,400	60	0	0	0	0
Average number	1,400	60	0	0	0	0
Avg live weight (lbs)	1,400	1,450	0	0		

Predominant milk cow breed: Holstein

Average milk production: 86 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 39,433.67 *tons per reporting period*

Total nitrogen from manure: 514,854.04 *lbs per reporting period*

After ammonia losses (30% loss applied): 360,397.83 *lbs per reporting period*

Total phosphorus from manure: 87,254.42 *lbs per reporting period*

Total potassium from manure: 282,829.72 *lbs per reporting period*

Total salt from manure: 672,987.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 23,525,280 *gallons*

Total nitrogen generated: 134,215.71 *lbs*

Total phosphorus generated: 59,280.86 *lbs*

Total potassium generated: 79,444.04 *lbs*

Total salt generated: 1,363,826.57 *lbs*

	23,525,280	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	23,525,280	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
MID Lateral 3	Surface water
TID Lateral 3	Surface water
W10-41 Dakota IW	Ground water
W1-1-Dairy 1 IW	Ground water
W9-11 McCarthy IW	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/15/2023	Solid commercial fertilizer *UN32*	21.94 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	14,027.56	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	14,027.56	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Almonds	47	47	1	none	X017-X062-X010-XXXX
Boland	36	36	2	process wastewater	X017-X062-X044-XXXX
Borland	37	37	2	process wastewater	X022-X002-X026-XXXX
Dairy 1	54	54	2	process wastewater	X022-X002-X062-XXXX X022-X002-X063-XXXX
Dakota	54	54	1	none	X076-X026-X003-XXXX
Harland	36	36	2	process wastewater	X022-X003-X005-XXXX
McCarthy	108	108	1	none	X017-X063-X026-XXXX X017-X063-X027-XXXX
Rogers	75	75	2	process wastewater	X017-X062-X010-XXXX X017-X062-X037-XXXX
Zwart	37	37	2	process wastewater	X017-X062-X035-XXXX
Totals for areas that were used for application	275	275	12		
Totals for areas that were not used for application	209	209	3		
Land application area totals	484	484	15		

B. CROPS AND HARVESTS

Almonds

Field name: Almonds

01/01/2022: Almond, in shell

Crop: Almond, in shell Acres planted: 33 Plant date: 01/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	150.00 *ton*	As-is		17.8	6,580.00	1,010.00	30,200.00		9.83

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	4.55	59.82	9.18	274.55	734.57

Boland

Boland

Field name: Boland

10/29/2002: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 36 Plant date: 10/29/2002

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	764.00 *ton*	As-is		70.0	3,600.00	840.00	7,990.00		9.58

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	21.22	152.80	35.65	339.13	1,219.85

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	1,092.00 *ton*	As-is		68.4	3,030.00	360.00	5,680.00		6.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.33	183.82	21.84	344.59	1,150.24

Borland

Field name: Borland

10/29/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 37 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	596.00 *ton*	As-is		70.0	6,160.00	1,380.00	110,000.00		10.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	16.11	198.45	44.46	3,543.78	1,043.81

Borland

05/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 37 Plant date: 05/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/09/2023	1,037.00 *ton*	As-is		68.7	3,570.00	620.00	4,140.00		4.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.03	200.11	34.75	232.06	842.16

Dairy 1

Field name: Dairy 1

10/31/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 54 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	1,031.00 *ton*	As-is		70.0	4,760.00	1,080.00	7,470.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	19.09	181.76	41.24	285.24	1,019.54

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 54 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	1,454.00 *ton*	As-is		62.2	4,610.00	560.00	4,930.00		5.99

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	26.93	248.26	30.16	265.49	1,219.32

Dakota

Field name: Dakota

01/01/2022: Almond, in shell

Crop: Almond, in shell Acres planted: 54 Plant date: 01/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	240.00 *ton*	As-is		17.8	6,580.00	1,010.00	30,200.00		9.83

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	4.44	58.49	8.98	268.44	718.25

Harland

Field name: Harland

10/31/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 36 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/01/2023	628.00 *ton*	As-is		70.0	5,290.00	1,020.00	8,270.00		9.77

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	17.44	184.56	35.59	288.53	1,022.59

05/29/2023: Corn, silage

Crop: Corn, silage Acres planted: 36 Plant date: 05/29/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/16/2023	1,103.00 *ton*	As-is		70.2	3,780.00	500.00	5,740.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.64	231.63	30.64	351.73	1,314.78

McCarthy

Field name: McCarthy

01/01/2022: Almond, in shell

Crop: Almond, in shell Acres planted: 108 Plant date: 01/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/21/2023	475.00 *ton*	As-is		17.8	6,580.00	1,010.00	30,200.00		9.83

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	4.00	172.00	29.20	186.00	0.00
Total actual harvest content	4.40	57.88	8.88	265.65	710.76

Rogers

Field name: Rogers

10/28/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 75 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	1,353.00 *ton*	As-is		70.0	5,140.00	1,130.00	11,100.00		8.15

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	18.04	185.45	40.77	400.49	882.16

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 75 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/11/2023	2,256.00 *ton*	As-is		72.1	3,260.00	530.00	3,750.00		6.19

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.08	196.12	31.88	225.60	1,038.97

Zwart

Field name: Zwart

10/28/2022: Triticale, soft dough

Crop: Triticale, soft dough Acres planted: 32 Plant date: 10/28/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/25/2023	604.00 *ton*	As-is		70.0	7,870.00	1,070.00	9,730.00		8.82

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	27.20	120.00	0.00
Total actual harvest content	18.88	297.09	40.39	367.31	998.87

05/26/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 05/26/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/14/2023	1,003.00 *ton*	As-is		66.2	3,450.00	530.00	5,010.00		5.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	31.34	216.27	33.22	314.06	1,101.80

NUTRIENT BUDGET

A. LAND APPLICATIONS

Almonds - 01/01/2022: Almond, in shell

Field name: Almonds

Crop: Almond, in shell

Plant date: 01/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	19.19	2,811,240.00 *gal*
Application event totals		0.00	0.00	0.00	19.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	24.00	0.00	0.00	0.00	
Application event totals		24.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	23.99	3,514,050.00 *gal*
Application event totals		0.00	0.00	0.00	23.99	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	24.00	0.00	0.00	0.00	
Application event totals		24.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	26.39	3,865,455.00 *gal*
Application event totals		0.00	0.00	0.00	26.39	

Almonds - 01/01/2022: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	21.59	3,162,645.00 *gal*
Application event totals		0.00	0.00	0.00	21.59	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	24.00	0.00	0.00	0.00	
Application event totals		24.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	125.96	18,448,763.00 *gal*
Application event totals		0.00	0.00	0.00	125.96	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	44.39	6,500,993.00 *gal*
Application event totals		0.00	0.00	0.00	44.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	28.79	4,216,860.00 *gal*
Application event totals		0.00	0.00	0.00	28.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	48.72	7,135,290.00 *gal*
Application event totals		0.00	0.00	0.00	48.72	

Boland - 10/29/2002: Triticale, soft dough

Boland - 10/29/2002: Triticale, soft dough

Field name: Boland

Crop: Triticale, soft dough

Plant date: 10/29/2002

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.20	47.50	0.72	424.34	405,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	27.20	4,345,650.00 *gal*
Application event totals		55.20	47.50	0.72	451.54	

02/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.38	3.40	67.89	751.05	324,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	21.76	3,476,520.00 *gal*
Application event totals		60.38	3.40	67.89	772.81	

03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.51	4.54	90.53	1,001.40	432,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	25.39	4,055,940.00 *gal*
Application event totals		80.51	4.54	90.53	1,026.79	

Boland - 05/27/2023: Corn, silage

Field name: Boland

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.58	5.10	101.84	1,126.58	486,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	32.64	5,214,780.00 *gal*
Application event totals		90.58	5.10	101.84	1,159.21	

Boland - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	45.33	7,242,750.00 *gal*
Application event totals		0.00	0.00	0.00	45.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	58.88	50.67	0.77	452.63	432,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	29.01	4,635,360.00 *gal*
Application event totals		58.88	50.67	0.77	481.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	36.26	5,794,200.00 *gal*
Application event totals		0.00	0.00	0.00	36.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	51.52	44.34	0.68	396.05	378,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	25.39	4,055,940.00 *gal*
Application event totals		51.52	44.34	0.68	421.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	55.20	47.50	0.72	424.34	405,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	27.20	4,345,650.00 *gal*
Application event totals		55.20	47.50	0.72	451.54	

Boland - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	25.09	4,009,586.00 *gal*
Application event totals		0.00	0.00	0.00	25.09	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	25.39	4,055,940.00 *gal*
Application event totals		0.00	0.00	0.00	25.39	

Borland - 10/29/2022: Triticale, soft dough

Field name: Borland

Crop: Triticale, soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	71.61	61.63	0.94	550.50	540,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	26.84	4,408,020.00 *gal*
Application event totals		71.61	61.63	0.94	577.34	

02/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	81.60	4.60	91.75	1,014.93	450,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	22.20	3,645,000.00 *gal*
Application event totals		81.60	4.60	91.75	1,037.13	

Borland - 10/29/2022: Triticale, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	97.92	5.52	110.10	1,217.92	540,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	26.64	4,374,000.00 *gal*
Application event totals		97.92	5.52	110.10	1,244.55	

Borland - 05/24/2023: Corn, silage

Field name: Borland

Crop: Corn, silage

Plant date: 05/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	103.36	5.82	116.22	1,285.58	570,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	28.68	4,709,340.00 *gal*
Application event totals		103.36	5.82	116.22	1,314.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	53.88	8,848,602.00 *gal*
Application event totals		0.00	0.00	0.00	53.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	69.04	11,337,116.00 *gal*
Application event totals		0.00	0.00	0.00	69.04	

Borland - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.47	52.04	0.79	464.87	456,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	22.94	3,767,472.00 *gal*
Application event totals		60.47	52.04	0.79	487.81	

08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	48.14	41.43	0.63	370.06	363,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	18.26	2,999,106.00 *gal*
Application event totals		48.14	41.43	0.63	388.32	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.62	53.89	0.82	481.38	472,200.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	23.76	3,901,316.00 *gal*
Application event totals		62.62	53.89	0.82	505.14	

08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	38.58	6,335,302.00 *gal*
Application event totals		0.00	0.00	0.00	38.58	

08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	40.75	6,692,220.00 *gal*
Application event totals		0.00	0.00	0.00	40.75	

Borland - 05/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	39.33	6,459,232.00 *gal*
Application event totals		0.00	0.00	0.00	39.33	

Dairy 1 - 10/31/2022: Triticale, soft dough

Field name: Dairy 1

Crop: Triticale, soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	70.33	60.52	0.92	540.64	774,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	30.35	7,273,665.00 *gal*
Application event totals		70.33	60.52	0.92	570.99	

02/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.62	3.53	70.41	778.87	504,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	16.40	3,931,200.00 *gal*
Application event totals		62.62	3.53	70.41	795.27	

04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	98.40	5.54	110.64	1,223.93	792,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	25.78	6,177,600.00 *gal*
Application event totals		98.40	5.54	110.64	1,249.71	

Dairy 1 - 05/29/2023: Corn, silage

Dairy 1 - 05/29/2023: Corn, silage

Field name: Dairy 1

Crop: Corn, silage

Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.64	5.67	113.16	1,251.75	810,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	28.99	6,946,763.00 *gal*
Application event totals		100.64	5.67	113.16	1,280.74	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	23.19	5,557,410.00 *gal*
Application event totals		0.00	0.00	0.00	23.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	44.44	10,651,703.00 *gal*
Application event totals		0.00	0.00	0.00	44.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.45	46.00	0.70	410.89	588,240.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	21.05	5,044,893.00 *gal*
Application event totals		53.45	46.00	0.70	431.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.79	54.89	0.84	490.35	702,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	25.12	6,020,528.00 *gal*
Application event totals		63.79	54.89	0.84	515.47	

Dairy 1 - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	32.85	7,872,998.00 *gal*
Application event totals		0.00	0.00	0.00	32.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	53.45	46.00	0.70	410.89	588,240.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	21.05	5,044,893.00 *gal*
Application event totals		53.45	46.00	0.70	431.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	31.43	7,533,378.00 *gal*
Application event totals		0.00	0.00	0.00	31.43	

Dakota - 01/01/2022: Almond, in shell

Field name: Dakota

Crop: Almond, in shell

Plant date: 01/01/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	43.23	7,560,000.00 *gal*
Application event totals		0.00	0.00	0.00	43.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	24.00	0.00	0.00	0.00	
Application event totals		24.00	0.00	0.00	0.00	

Dakota - 01/01/2022: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	37.05	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.05	

05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	37.05	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.05	

05/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	24.00	0.00	0.00	0.00	
Application event totals		24.00	0.00	0.00	0.00	

05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	30.88	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.88	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	33.96	5,940,000.00 *gal*
Application event totals		0.00	0.00	0.00	33.96	

06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	24.00	0.00	0.00	0.00	
Application event totals		24.00	0.00	0.00	0.00	

Dakota - 01/01/2022: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	30.88	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.88	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	37.05	6,480,000.00 *gal*
Application event totals		0.00	0.00	0.00	37.05	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	30.88	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.88	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	24.70	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.70	

08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	24.70	4,320,000.00 *gal*
Application event totals		0.00	0.00	0.00	24.70	

08/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	30.88	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.88	

Dakota - 01/01/2022: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	30.88	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
MID Lateral 3	Surface water	0.00	0.00	0.00	30.88	5,400,000.00 *gal*
Application event totals		0.00	0.00	0.00	30.88	

Harland - 10/31/2022: Triticale, soft dough

Field name: Harland

Crop: Triticale, soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/09/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	85.87	73.89	1.13	660.09	630,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	28.69	4,584,195.00 *gal*
Application event totals		85.87	73.89	1.13	688.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.87	4.73	94.30	1,043.13	450,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	19.49	3,114,450.00 *gal*
Application event totals		83.87	4.73	94.30	1,062.62	

Harland - 10/31/2022: Triticale, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.28	4.41	88.01	973.58	420,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	18.19	2,906,820.00 *gal*
Application event totals		78.28	4.41	88.01	991.78	

Harland - 05/29/2023: Corn, silage

Field name: Harland

Crop: Corn, silage

Plant date: 05/29/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	115.74	6.52	130.13	1,439.51	621,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	26.67	4,261,613.00 *gal*
Application event totals		115.74	6.52	130.13	1,466.18	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	43.17	6,896,813.00 *gal*
Application event totals		0.00	0.00	0.00	43.17	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.39	69.18	1.06	617.97	589,800.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	25.33	4,047,503.00 *gal*
Application event totals		80.39	69.18	1.06	643.30	

Harland - 05/29/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	34.79	5,558,625.00 *gal*
Application event totals		0.00	0.00	0.00	34.79	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.88	58.41	0.89	521.79	498,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	21.39	3,417,525.00 *gal*
Application event totals		67.88	58.41	0.89	543.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.15	53.49	0.82	477.78	456,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	19.59	3,129,300.00 *gal*
Application event totals		62.15	53.49	0.82	497.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	25.44	4,063,973.00 *gal*
Application event totals		0.00	0.00	0.00	25.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	26.54	4,241,025.00 *gal*
Application event totals		0.00	0.00	0.00	26.54	

McCarthy - 01/01/2022: Almond, in shell

Field name: McCarthy

Crop: Almond, in shell

Plant date: 01/01/2022

McCarthy - 01/01/2022: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	24.00	0.00	0.00	0.00	
Application event totals		24.00	0.00	0.00	0.00	

04/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	9.51	4,556,790.00 *gal*
Application event totals		0.00	0.00	0.00	9.51	

05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	9.51	4,556,790.00 *gal*
Application event totals		0.00	0.00	0.00	9.51	

05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	10.19	4,882,275.00 *gal*
Application event totals		0.00	0.00	0.00	10.19	

05/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	24.00	0.00	0.00	0.00	
Application event totals		24.00	0.00	0.00	0.00	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	10.19	4,882,275.00 *gal*
Application event totals		0.00	0.00	0.00	10.19	

McCarthy - 01/01/2022: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	24.00	0.00	0.00	0.00	
Application event totals		24.00	0.00	0.00	0.00	

07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	8.83	4,231,305.00 *gal*
Application event totals		0.00	0.00	0.00	8.83	

07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	9.51	4,556,790.00 *gal*
Application event totals		0.00	0.00	0.00	9.51	

08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	9.51	4,556,790.00 *gal*
Application event totals		0.00	0.00	0.00	9.51	

09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	8.83	4,231,305.00 *gal*
Application event totals		0.00	0.00	0.00	8.83	

09/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	9.51	4,556,790.00 *gal*
Application event totals		0.00	0.00	0.00	9.51	

McCarthy - 01/01/2022: Almond, in shell

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	11.54	5,533,245.00 *gal*
Application event totals		0.00	0.00	0.00	11.54	

Rogers - 10/28/2022: Triticale, soft dough

Field name: Rogers

Crop: Triticale, soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/06/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	62.81	54.05	0.82	482.81	960,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	24.66	8,208,000.00 *gal*
Application event totals		62.81	54.05	0.82	507.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	96.62	5.44	108.63	1,201.68	1,080,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	14.60	4,860,000.00 *gal*
Application event totals		96.62	5.44	108.63	1,216.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.15	4.23	84.49	934.64	840,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	29.56	9,839,760.00 *gal*
Application event totals		75.15	4.23	84.49	964.20	

Rogers - 05/26/2023: Corn, silage

Rogers - 05/26/2023: Corn, silage

Field name: Rogers

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	67.09	3.78	75.44	834.50	750,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	26.39	8,785,125.00 *gal*
Application event totals		67.09	3.78	75.44	860.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	32.73	10,893,555.00 *gal*
Application event totals		0.00	0.00	0.00	32.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	52.99	45.60	0.70	407.37	810,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	28.50	9,487,935.00 *gal*
Application event totals		52.99	45.60	0.70	435.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	33.78	11,244,960.00 *gal*
Application event totals		0.00	0.00	0.00	33.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	35.33	30.40	0.46	271.58	540,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	19.00	6,325,290.00 *gal*
Application event totals		35.33	30.40	0.46	290.58	

Rogers - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	54.88	47.23	0.72	421.85	838,800.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	29.52	9,825,284.00 *gal*
Application event totals		54.88	47.23	0.72	451.37	

08/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	16.89	5,622,480.00 *gal*
Application event totals		0.00	0.00	0.00	16.89	

09/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	23.23	7,730,910.00 *gal*
Application event totals		0.00	0.00	0.00	23.23	

Zwart - 10/28/2022: Triticale, soft dough

Field name: Zwart

Crop: Triticale, soft dough

Plant date: 10/28/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/07/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	73.60	63.34	0.97	565.79	480,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	30.72	4,363,200.00 *gal*
Application event totals		73.60	63.34	0.97	596.51	

Zwart - 10/28/2022: Triticale, soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	75.48	4.25	84.87	938.81	360,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	23.04	3,272,400.00 *gal*
Application event totals		75.48	4.25	84.87	961.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	100.64	5.67	113.16	1,251.75	480,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	30.72	4,363,200.00 *gal*
Application event totals		100.64	5.67	113.16	1,282.47	

Zwart - 05/26/2023: Corn, silage

Field name: Zwart

Crop: Corn, silage

Plant date: 05/26/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	106.93	6.02	120.23	1,329.98	510,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	32.64	4,635,900.00 *gal*
Application event totals		106.93	6.02	120.23	1,362.63	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	30.72	4,363,200.00 *gal*
Application event totals		0.00	0.00	0.00	30.72	

Zwart - 05/26/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	42.24	5,999,400.00 *gal*
Application event totals		0.00	0.00	0.00	42.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.80	51.46	0.79	459.71	390,000.00 *gal*
Application event totals		59.80	51.46	0.79	459.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	64.40	55.42	0.85	495.07	420,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	26.88	3,817,800.00 *gal*
Application event totals		64.40	55.42	0.85	521.95	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	30.72	4,363,200.00 *gal*
Application event totals		0.00	0.00	0.00	30.72	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	59.80	51.46	0.79	459.71	390,000.00 *gal*
TID Lateral 3	Surface water	0.00	0.00	0.00	24.96	3,545,100.00 *gal*
Application event totals		59.80	51.46	0.79	484.67	

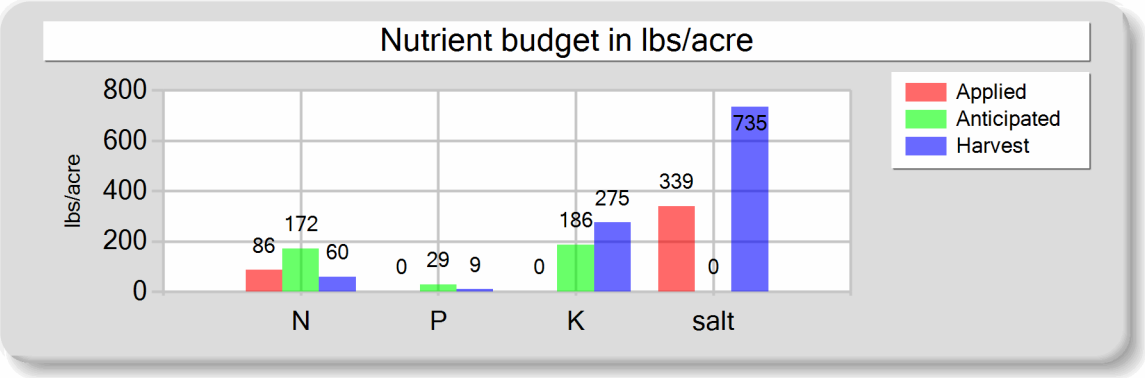
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
TID Lateral 3	Surface water	0.00	0.00	0.00	23.04	3,272,400.00 *gal*
Application event totals		0.00	0.00	0.00	23.04	

B. NUTRIENT BUDGET

Almonds - 01/01/2022: Almond, in shell

Field name: Almonds Crop: Almond, in shell Plant date: 01/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	72.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	339.03
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	86.00	0.00	0.00	339.03
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	59.82	9.18	274.55	734.57
Nutrient balance	26.18	-9.18	-274.55	-395.54
Applied to removed ratio	1.44	0.00	0.00	0.46

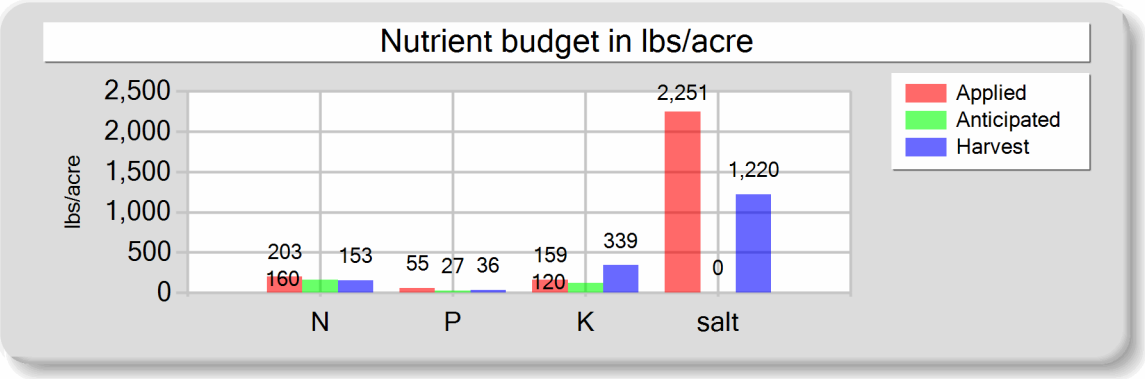
Fresh water applied	
49,655,296.00	*gallons*
1,828.64	*acre-inches*
55.41	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Boland - 10/29/2002: Triticale, soft dough

Field name: Boland | Crop: Triticale, soft dough | Plant date: 10/29/2002



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	196.10	55.44	159.15	2,176.79
Fresh water	0.00	0.00	0.00	74.34
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	203.10	55.44	159.15	2,251.14
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	152.80	35.65	339.13	1,219.85
Nutrient balance	50.30	19.79	-179.98	1,031.28
Applied to removed ratio	1.33	1.56	0.47	1.85

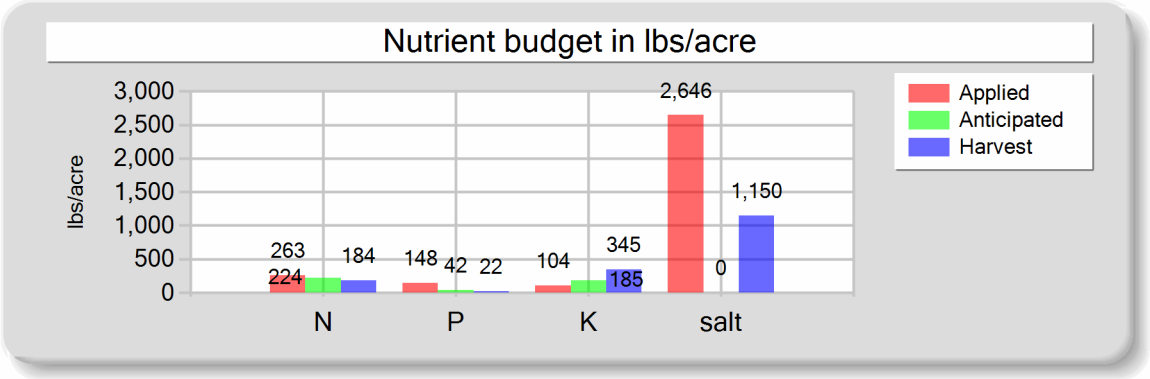
Fresh water applied	
11,878,110.00	*gallons*
437.43	*acre-inches*
12.15	*inches/acre*

Process wastewater applied	
1,161,000.00	*gallons*
42.76	*acre-inches*
1.19	*inches/acre*

Total harvests for the crop	
1	*harvests*

Boland - 05/27/2023: Corn, silage

Field name: Boland | Crop: Corn, silage | Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	256.18	147.62	104.02	2,399.60
Fresh water	0.00	0.00	0.00	246.31
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	263.18	147.62	104.02	2,645.91
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	183.82	21.84	344.59	1,150.24
Nutrient balance	79.36	125.78	-240.57	1,495.67
Applied to removed ratio	1.43	6.76	0.30	2.30

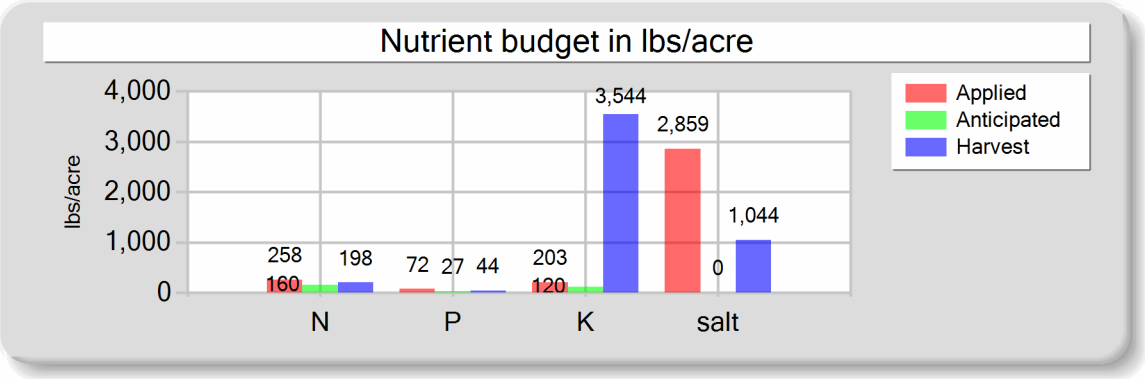
Fresh water applied	
39,354,206.00	*gallons*
1,449.28	*acre-inches*
40.26	*inches/acre*

Process wastewater applied	
1,701,000.00	*gallons*
62.64	*acre-inches*
1.74	*inches/acre*

Total harvests for the crop	
1	*harvests*

Borland - 10/29/2022: Triticale, soft dough

Field name: Borland | Crop: Triticale, soft dough | Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	251.13	71.74	202.79	2,783.35
Fresh water	0.00	0.00	0.00	75.68
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	258.13	71.74	202.79	2,859.03
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	198.45	44.46	3,543.78	1,043.81
Nutrient balance	59.68	27.28	-3,340.99	1,815.22
Applied to removed ratio	1.30	1.61	0.06	2.74

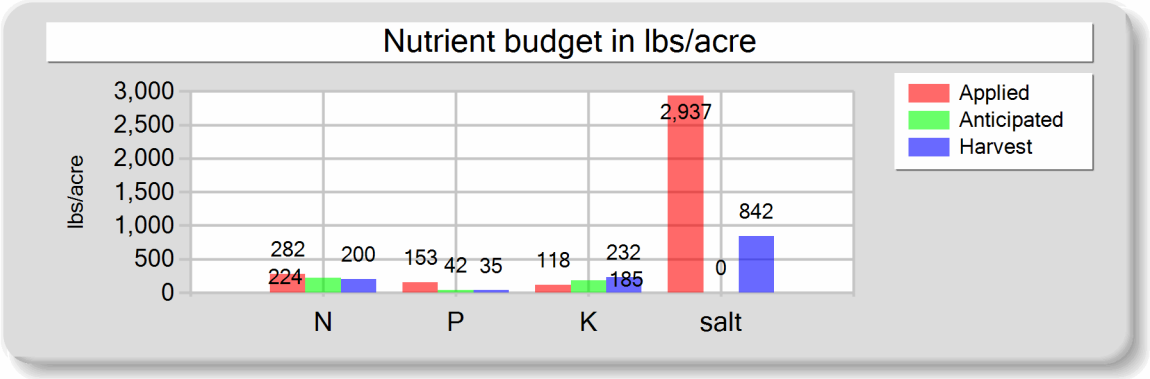
Fresh water applied	
12,427,020.00	*gallons*
457.64	*acre-inches*
12.37	*inches/acre*

Process wastewater applied	
1,530,000.00	*gallons*
56.34	*acre-inches*
1.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

Borland - 05/24/2023: Corn, silage

Field name: Borland | Crop: Corn, silage | Plant date: 05/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	274.60	153.18	118.46	2,601.89
Fresh water	0.00	0.00	0.00	335.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	281.60	153.18	118.46	2,937.12
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	200.11	34.75	232.06	842.16
Nutrient balance	81.48	118.43	-113.60	2,094.96
Applied to removed ratio	1.41	4.41	0.51	3.49

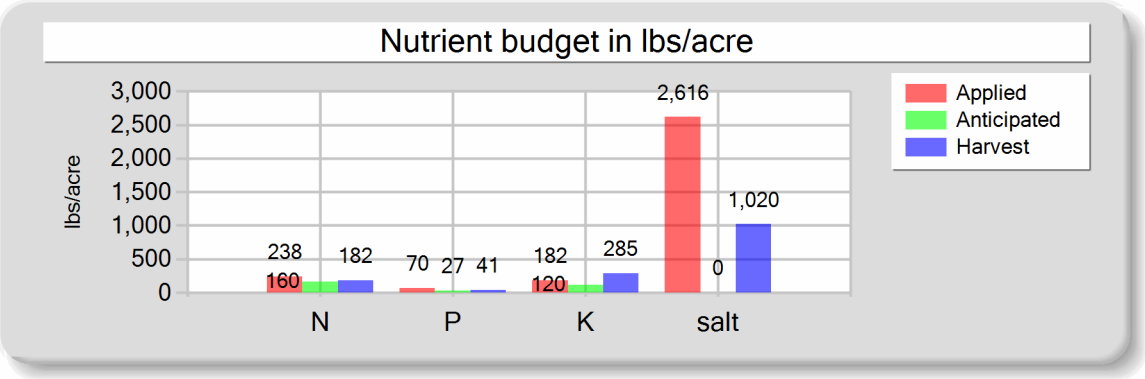
Fresh water applied	
55,049,706.00	*gallons*
2,027.29	*acre-inches*
54.79	*inches/acre*

Process wastewater applied	
1,861,200.00	*gallons*
68.54	*acre-inches*
1.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 1 - 10/31/2022: Triticale, soft dough

Field name: Dairy 1 Crop: Triticale, soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	231.36	69.60	181.98	2,543.44
Fresh water	0.00	0.00	0.00	72.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.36	69.60	181.98	2,615.97
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	181.76	41.24	285.24	1,019.54
Nutrient balance	56.60	28.36	-103.27	1,596.43
Applied to removed ratio	1.31	1.69	0.64	2.57

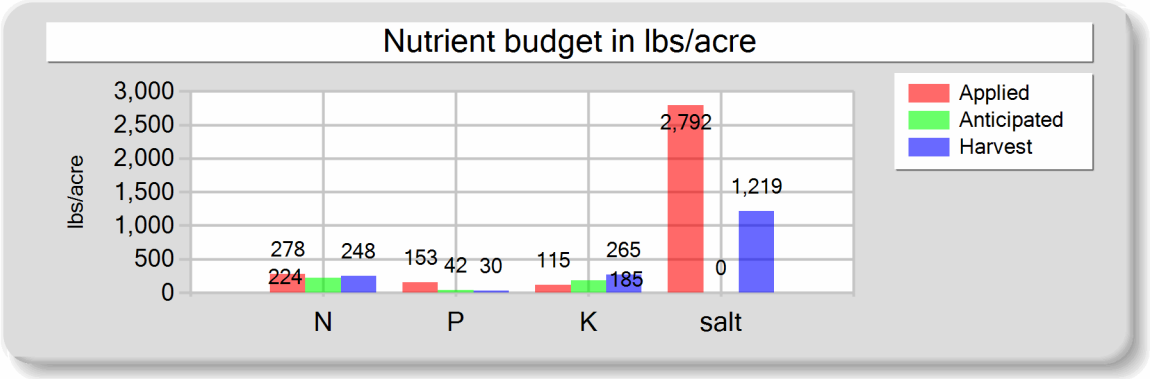
Fresh water applied	
17,382,465.00	*gallons*
640.14	*acre-inches*
11.85	*inches/acre*

Process wastewater applied	
2,070,000.00	*gallons*
76.23	*acre-inches*
1.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dairy 1 - 05/29/2023: Corn, silage

Field name: Dairy 1 Crop: Corn, silage Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	271.33	152.56	115.40	2,563.88
Fresh water	0.00	0.00	0.00	228.12
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	278.33	152.56	115.40	2,792.00
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	248.26	30.16	265.49	1,219.32
Nutrient balance	30.08	122.40	-150.09	1,572.68
Applied to removed ratio	1.12	5.06	0.43	2.29

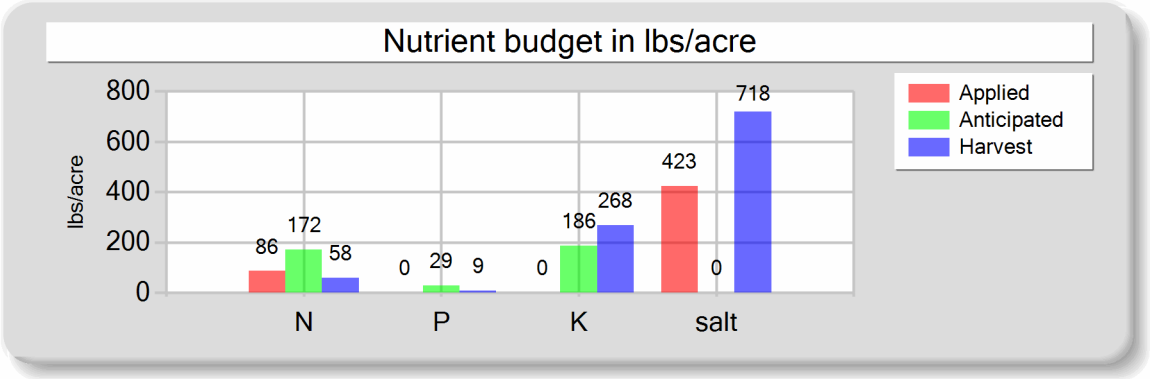
Fresh water applied	
54,672,566.00	*gallons*
2,013.40	*acre-inches*
37.29	*inches/acre*

Process wastewater applied	
2,688,480.00	*gallons*
99.01	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

Dakota - 01/01/2022: Almond, in shell

Field name: Dakota | Crop: Almond, in shell | Plant date: 01/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	72.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	423.01
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	86.00	0.00	0.00	423.01
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	58.49	8.98	268.44	718.25
Nutrient balance	27.51	-8.98	-268.44	-295.24
Applied to removed ratio	1.47	0.00	0.00	0.59

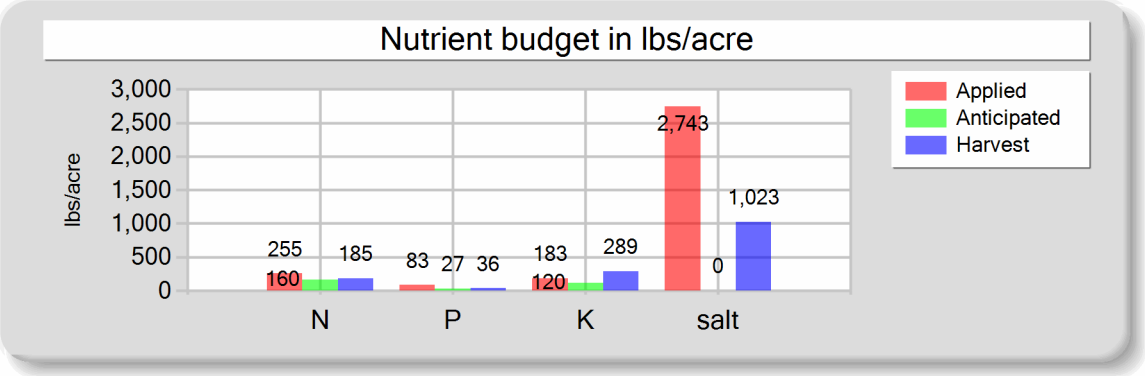
Fresh water applied	
73,980,000.00	*gallons*
2,724.43	*acre-inches*
50.45	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Harland - 10/31/2022: Triticale, soft dough

Field name: Harland | Crop: Triticale, soft dough | Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	248.01	83.03	183.44	2,676.80
Fresh water	0.00	0.00	0.00	66.38
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.01	83.03	183.44	2,743.17
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	184.56	35.59	288.53	1,022.59
Nutrient balance	70.45	47.44	-105.09	1,720.58
Applied to removed ratio	1.38	2.33	0.64	2.68

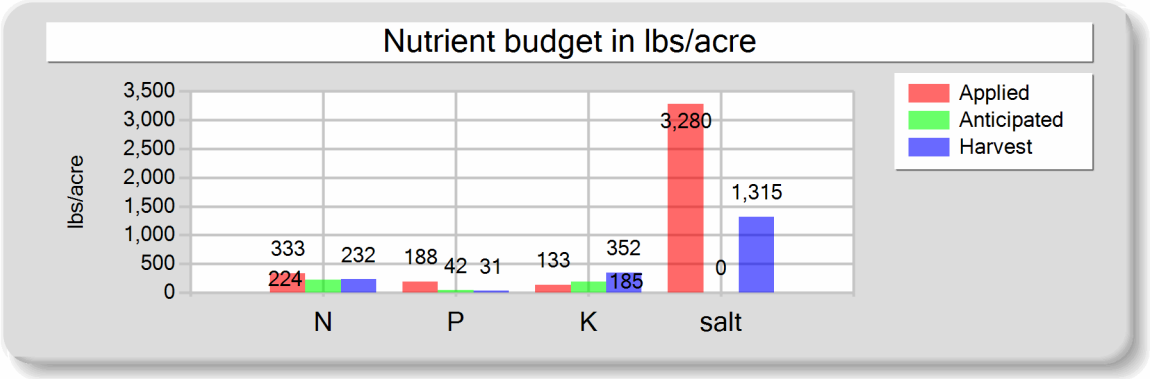
Fresh water applied	
10,605,465.00	*gallons*
390.56	*acre-inches*
10.85	*inches/acre*

Process wastewater applied	
1,500,000.00	*gallons*
55.24	*acre-inches*
1.53	*inches/acre*

Total harvests for the crop	
1	*harvests*

Harland - 05/29/2023: Corn, silage

Field name: Harland | Crop: Corn, silage | Plant date: 05/29/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	326.16	187.60	132.89	3,057.05
Fresh water	0.00	0.00	0.00	222.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	333.16	187.60	132.89	3,279.96
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	231.63	30.64	351.73	1,314.78
Nutrient balance	101.53	156.96	-218.84	1,965.18
Applied to removed ratio	1.44	6.12	0.38	2.49

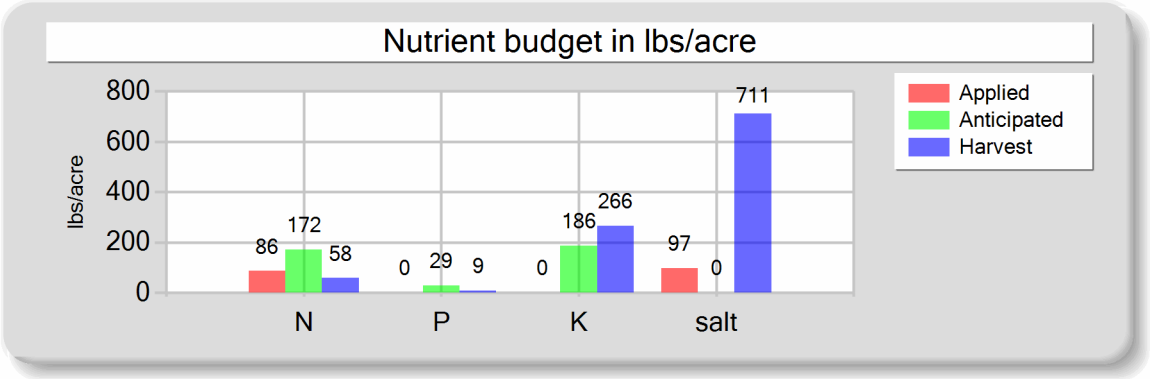
Fresh water applied	
35,616,377.00	*gallons*
1,311.63	*acre-inches*
36.43	*inches/acre*

Process wastewater applied	
2,164,800.00	*gallons*
79.72	*acre-inches*
2.21	*inches/acre*

Total harvests for the crop	
1	*harvests*

McCarthy - 01/01/2022: Almond, in shell

Field name: McCarthy | Crop: Almond, in shell | Plant date: 01/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	72.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	97.10
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	86.00	0.00	0.00	97.10
Anticipated crop nutrient removal	172.00	29.20	186.00	0.00
Actual crop nutrient removal	57.88	8.88	265.65	710.76
Nutrient balance	28.12	-8.88	-265.65	-613.66
Applied to removed ratio	1.49	0.00	0.00	0.14

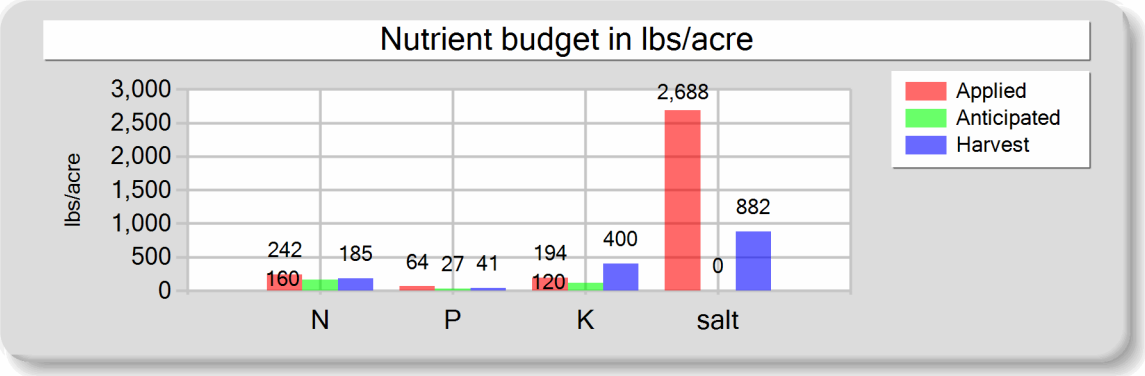
Fresh water applied	
46,544,355.00	*gallons*
1,714.07	*acre-inches*
15.87	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Rogers - 10/28/2022: Triticale, soft dough

Field name: Rogers | Crop: Triticale, soft dough | Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	234.57	63.73	193.95	2,619.13
Fresh water	0.00	0.00	0.00	68.82
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	241.57	63.73	193.95	2,687.95
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	185.45	40.77	400.49	882.16
Nutrient balance	56.12	22.96	-206.54	1,805.79
Applied to removed ratio	1.30	1.56	0.48	3.05

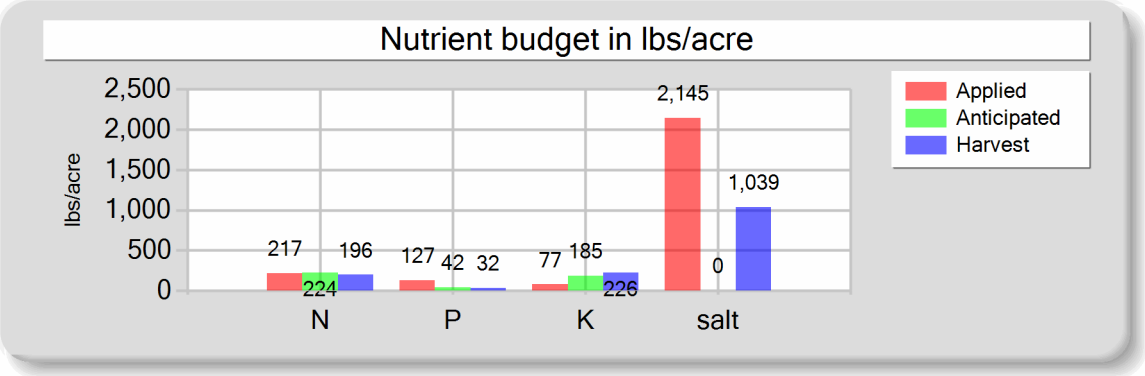
Fresh water applied	
22,907,760.00	*gallons*
843.61	*acre-inches*
11.25	*inches/acre*

Process wastewater applied	
2,880,000.00	*gallons*
106.06	*acre-inches*
1.41	*inches/acre*

Total harvests for the crop	
1	*harvests*

Rogers - 05/26/2023: Corn, silage

Field name: Rogers Crop: Corn, silage Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	210.30	127.01	77.32	1,935.30
Fresh water	0.00	0.00	0.00	210.04
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	217.30	127.01	77.32	2,145.34
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	196.12	31.88	225.60	1,038.97
Nutrient balance	21.17	95.13	-148.28	1,106.37
Applied to removed ratio	1.11	3.98	0.34	2.06

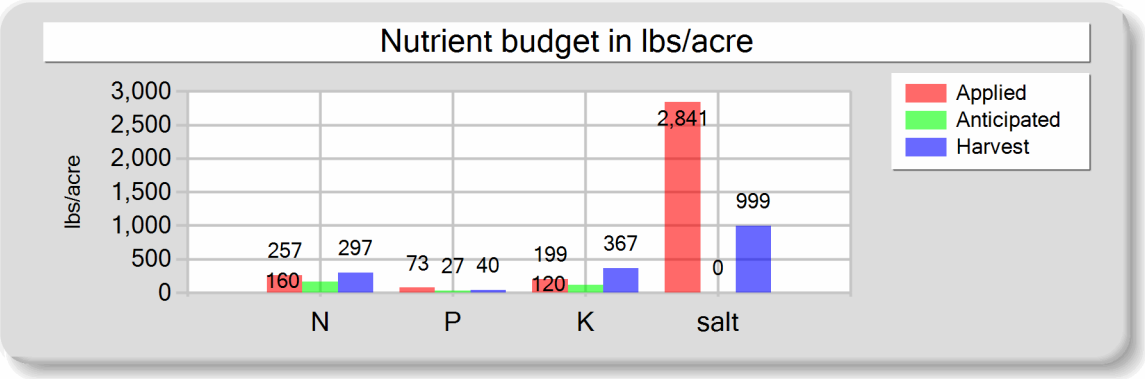
Fresh water applied	
69,915,539.00	*gallons*
2,574.75	*acre-inches*
34.33	*inches/acre*

Process wastewater applied	
2,938,800.00	*gallons*
108.23	*acre-inches*
1.44	*inches/acre*

Total harvests for the crop	
1	*harvests*

Zwart - 10/28/2022: Triticale, soft dough

Field name: Zwart | Crop: Triticale, soft dough | Plant date: 10/28/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	249.72	73.26	198.99	2,756.35
Fresh water	0.00	0.00	0.00	84.48
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	256.72	73.26	198.99	2,840.84
Anticipated crop nutrient removal	160.00	27.20	120.00	0.00
Actual crop nutrient removal	297.09	40.39	367.31	998.87
Nutrient balance	-40.37	32.87	-168.31	1,841.97
Applied to removed ratio	0.86	1.81	0.54	2.84

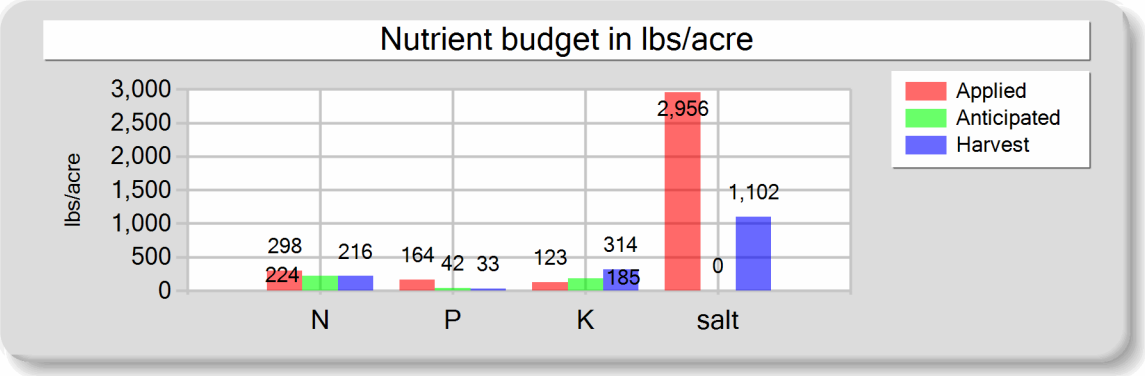
Fresh water applied	
11,998,800.00	*gallons*
441.88	*acre-inches*
13.81	*inches/acre*

Process wastewater applied	
1,320,000.00	*gallons*
48.61	*acre-inches*
1.52	*inches/acre*

Total harvests for the crop	
1	*harvests*

Zwart - 05/26/2023: Corn, silage

Field name: Zwart | Crop: Corn, silage | Plant date: 05/26/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	290.94	164.37	122.65	2,744.46
Fresh water	0.00	0.00	0.00	211.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	297.94	164.37	122.65	2,955.67
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	216.27	33.22	314.06	1,101.80
Nutrient balance	81.67	131.15	-191.42	1,853.88
Applied to removed ratio	1.38	4.95	0.39	2.68

Fresh water applied	
29,997,000.00	*gallons*
1,104.69	*acre-inches*
34.52	*inches/acre*

Process wastewater applied	
1,710,000.00	*gallons*
62.97	*acre-inches*
1.97	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/03/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 11.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	41,000.00	26,300.00	17,500.00	33,000.00	29,300.00	5,830.00	7,420.00	2,220.00		36.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/03/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 54.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	16,200.00	7,110.00	9,410.00	22,200.00	11,100.00	2,210.00	3,920.00	1,400.00		71.90
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWS

Sample and source description: WWS

Sample date: 05/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,080.00	442.00	0.00	0.00	97.10	861.00	285.00	206.00	343.00	4,760.00	0.00	496.00	357.00	8,640.00	4,760
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWS

Sample and source description: WWS

Sample date: 05/13/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	893.00	484.00	0.00	0.00	71.70	874.00	280.00	190.00	346.00	4,360.00	0.00	437.00	423.00	8,620.00	4,660
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWS

Sample and source description: WWS

Sample date: 07/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	588.00	386.00	0.00	0.00	506.00	7.72								7,790.00	4,520
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWS

Sample and source description: WWS

Sample date: 07/22/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	430.00	295.00	0.00	0.00	19.70	342.00								6,850.00	4,220
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWS

Sample and source description: WWS

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	482.00	314.00	0.00	0.00	35.60	596.00								6,210.00	7,400
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWS

Sample and source description: WWS

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	409.00	272.00	0.00	0.00	34.40	504.00								5,380.00	5,160
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWS

Sample and source description: WWS

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	375.00	246.00	0.00	0.00	35.40	491.00								4,970.00	4,400
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWS

Sample and source description: WWS

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1,640.00	619.00	0.00	0.00	52.10	1,110.00								10,800.00	7,340
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWS

Sample and source description: WWS

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	804.00	538.00	0.00	0.00	45.30	904.00								9,370.00	10,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

WWS

Sample and source description: WWS

Sample date: 05/16/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	815.00	496.00	0.00	0.00	42.50	838.00								8,720.00	9,640
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

MID Lateral 3

MID Lateral 3

Sample description: MID Lateral 3

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								30.70	37
DL	1.00	0.50	0.50								10.00	10

TID Lateral 3

TID Lateral 3

Sample description: TID Lateral 3

Sample date: 07/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								35.90	27
DL	1.00	0.50	0.50								10.00	10

W10-41 Dakota IW

W10-41 Dakota IW

W10-41 Dakota IW

Sample description: W10-41 Dakota IW

Sample date: 12/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	21.10	0.00	18.30								918.00	555
DL	0.50	0.50	0.50								1.00	10

W9-11 McCarthy IW

W9-11 McCarthy IW

Sample description: W9-11 McCarthy IW

Sample date: 12/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	25.90	0.00	0.00								1,070.00	545
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Almonds - 01/01/2022: Almond, in shell

Almonds - 01/01/2022: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 17.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,580.00	1,010.00	30,200.00		9.83
DL	100.00	100.00	100.00		0.01

Boland - 10/29/2002: Triticale, soft dough

Triticale Silage

Sample and source description: Triticale Silage

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,600.00	840.00	7,990.00		9.58
DL	100.00	100.00	100.00		0.01

Boland - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,030.00	360.00	5,680.00		6.00
DL	100.00	100.00	100.00		0.01

Borland - 10/29/2022: Triticale, soft dough

Triticale Silage

Sample and source description: Triticale Silage

Sample date: 04/26/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,160.00	1,380.00	11,000.00		10.80
DL	100.00	100.00	100.00		0.01

Borland - 05/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,570.00	620.00	4,140.00		4.80
DL	100.00	100.00	100.00		0.01

Dairy 1 - 10/31/2022: Triticale, soft dough

Triticale Silage

Sample and source description: Triticale Silage

Sample date: 05/08/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,780.00	1,080.00	7,470.00		8.90
DL	100.00	100.00	100.00		0.01

Dairy 1 - 05/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 62.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,610.00	560.00	4,930.00		5.99
DL	100.00	100.00	100.00		0.01

Dakota - 01/01/2022: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 17.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,580.00	1,010.00	30,200.00		9.83
DL	100.00	100.00	100.00		0.01

Harland - 10/31/2022: Triticale, soft dough

Triticale Silage

Sample and source description: Triticale Silage

Sample date: 05/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,290.00	1,020.00	8,270.00		9.77
DL	100.00	100.00	100.00		0.01

Harland - 05/29/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,780.00	500.00	5,740.00		7.20
DL	100.00	100.00	100.00		0.01

McCarthy - 01/01/2022: Almond, in shell

Almonds

Sample and source description: Almonds

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 17.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,580.00	1,010.00	30,200.00		9.83
DL	100.00	100.00	100.00		0.01

Rogers - 10/28/2022: Triticale, soft dough

Triticale

Sample and source description: Triticale

Sample date: 04/28/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,140.00	1,130.00	11,100.00		8.15
DL	100.00	100.00	100.00		0.01

Rogers - 05/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,260.00	530.00	3,750.00		6.19
DL	100.00	100.00	100.00		0.01

Zwart - 10/28/2022: Triticale, soft dough

Triticale Silage

Sample and source description: Triticale Silage

Sample date: 04/27/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,870.00	1,070.00	9,730.00		8.82
DL	100.00	100.00	100.00		0.01

Zwart - 05/26/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/15/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 66.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,450.00	530.00	5,010.00		5.20
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

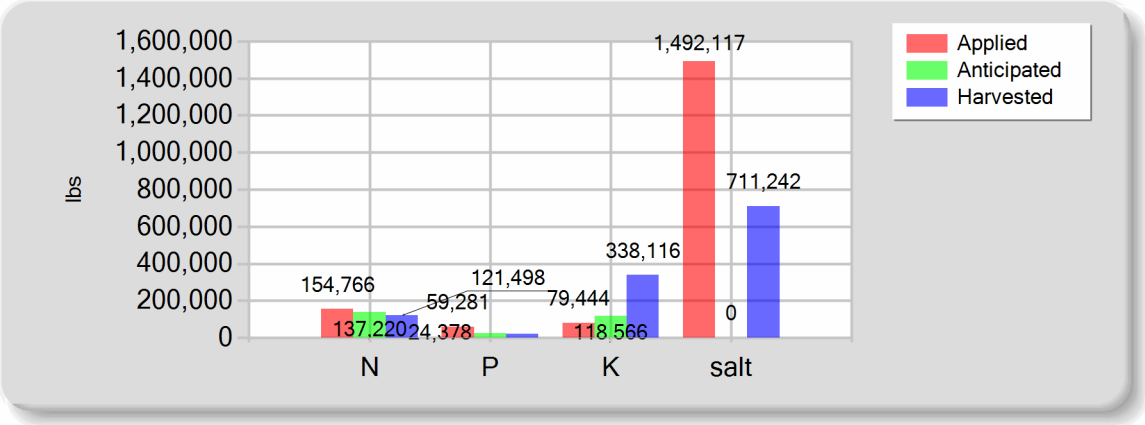
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

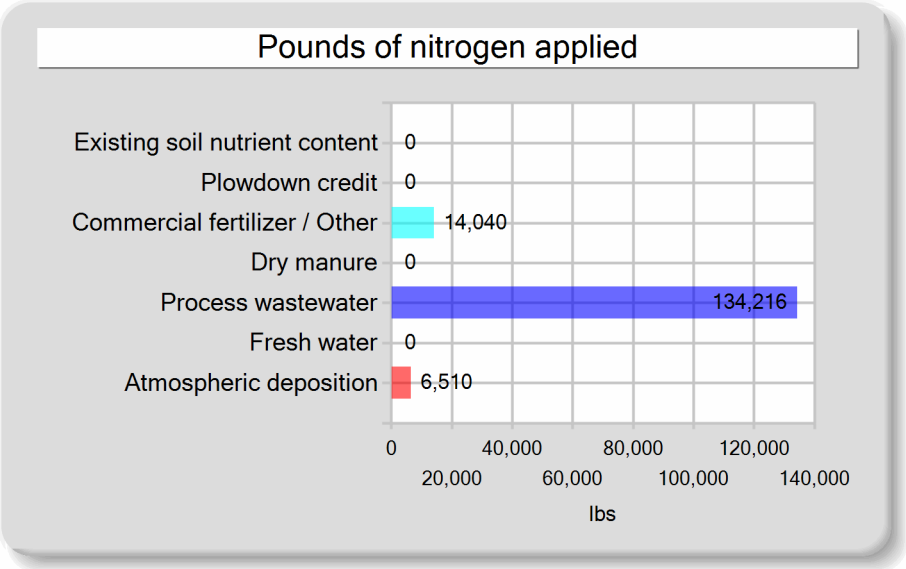
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

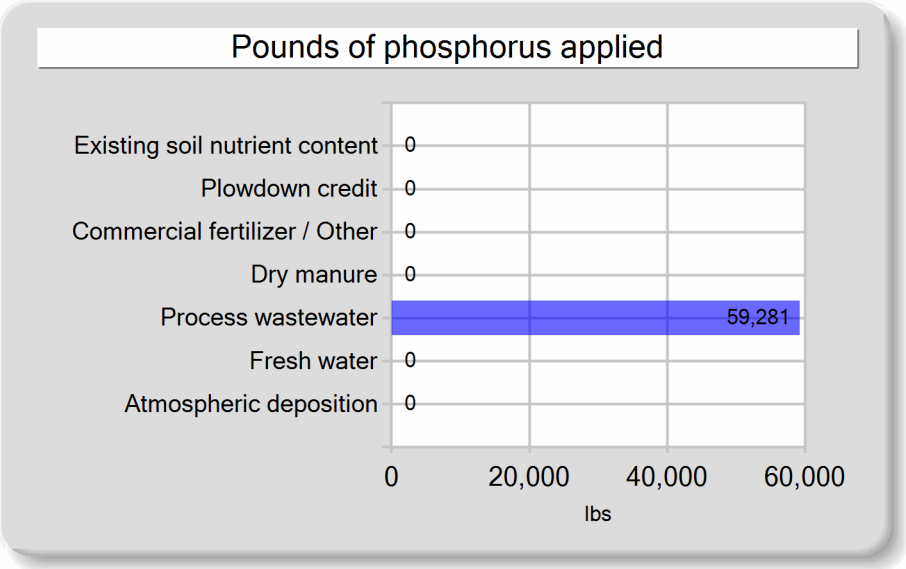
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	14,040.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	134,215.71	59,280.86	79,444.04	1,363,826.57
Fresh water	0.00	0.00	0.00	128,290.91
Atmospheric deposition	6,510.00	0.00	0.00	0.00
Total nutrients applied	154,765.71	59,280.86	79,444.04	1,492,117.47
Anticipated crop nutrient removal	137,220.00	24,378.00	118,566.00	0.00
Actual crop nutrient removal	121,498.16	20,792.34	338,115.84	711,242.17
Nutrient balance	33,267.55	38,488.52	-258,671.80	780,875.30
Applied to removed ratio	1.27	2.85	0.23	2.10

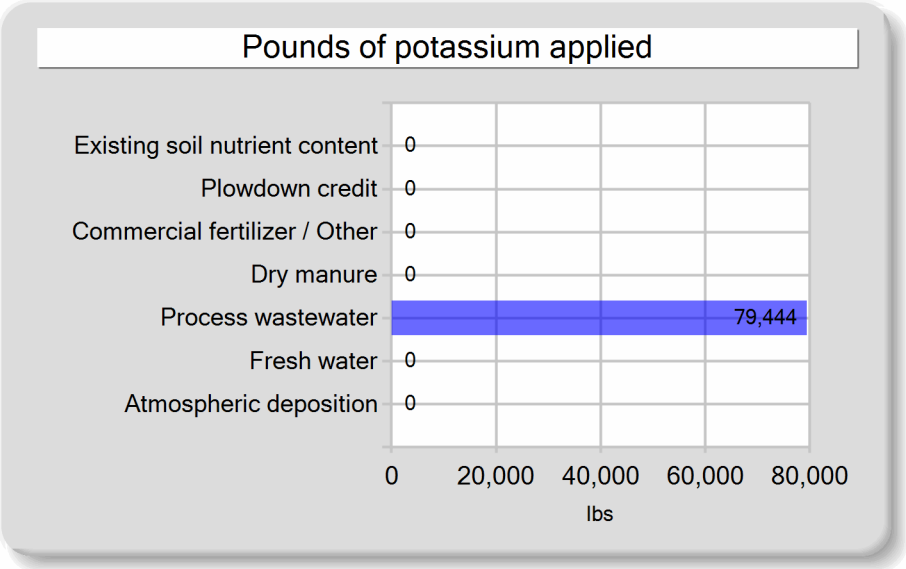
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

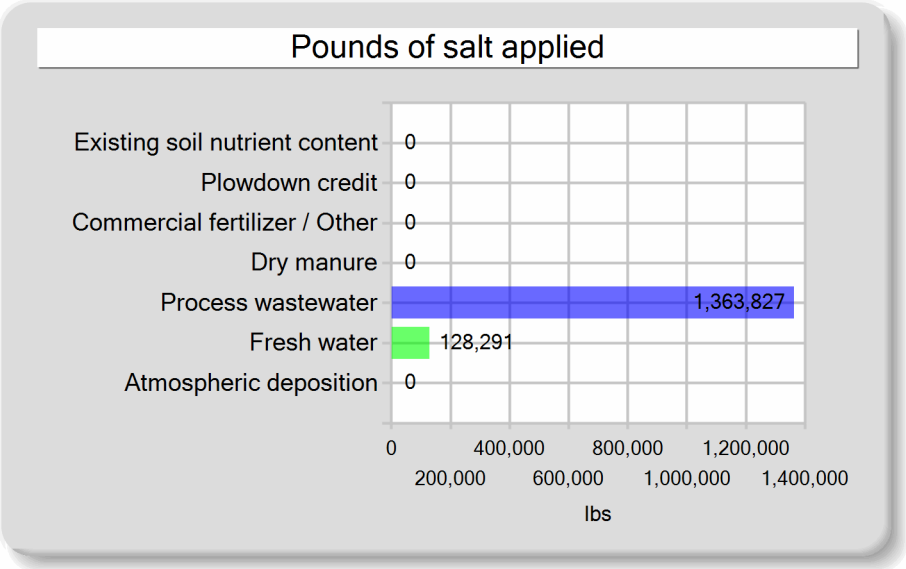


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	No
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	No

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

mike zylstra
mike zylstra (Jun 26, 2024 12:22 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Michael John Zylstra	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jun 26, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W6-1 Simmons	T3346003A	Electrical Conductivity (EC)	EPA 120.1 and	1.28	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	57.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	821	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

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SOURCE: Well Water

NOTES: Temperature was above qualifying at 17.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W7-1 3501 W. Monte Vista	T3346003B	Electrical Conductivity (EC)	EPA 120.1 and	1.44	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	58.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	922	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.8c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W8-1 6807 Jennings (House)	T3346003C	Electrical Conductivity (EC)	EPA 120.1 and	2.60	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	77.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1670	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 18.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W8-2 6807 Jennings (Mobile)	T3346003D	Electrical Conductivity (EC)	EPA 120.1 and	2.62	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	76.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1680	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: _____

Client: ANDREW ZYLSTRA DAIRY INC
Mailing address: 2655 W. Monte Vista Ave
(not facility address)
Phone#:

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

H1	H3	M3	F18	OTHER

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis				
1	12-12	10:53	W6-1 - Simmons		17.6		H1				
2	↓	↓	W7-1 - 3501 W Monte Vista		17.6		↓				
3			W8-1 - 6807 Jennings (House)		18.8						
4			W8-2 - 6807 Jennings (Mobile)		18.2						
5											
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	*(LABORATORY USE ONLY)* Logged in By:	LABORATORY NUMBER
1st	[signature]	Andrew Zylstra	[illegible]			
2nd	[signature]	DENELE ANALYTICAL, INC.	12-12-23 @ 11:59			T3346003

Required to be submitted to any agency? (YES / NO)

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SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 17.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W1-1 - Dairy 1 - Milk Barn	T3346002A	Electrical Conductivity (EC)	EPA 120.1 and	1.31	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	22.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	836	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	100	105 (T3347004A)	102 (T3347004A)	3.32

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Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W1-3 - 7800 Vivian	T3346002B	Electrical Conductivity (EC)	EPA 120.1 and	1.96	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	42.6	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1250	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	100	105 (T3347004A)	102 (T3347004A)	3.32

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Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W2-1 - Dairy 2-A	T3346002C	Electrical Conductivity (EC)	EPA 120.1 and	1.15	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	27.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	737	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	100	105 (T3347004A)	102 (T3347004A)	3.32

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Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W2-2 - Dairy 2-B	T3346002D	Electrical Conductivity (EC)	EPA 120.1 and	1.11	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	26.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	707	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 18.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W2-4 - 7855 S. Carpenter	T3346002E	Electrical Conductivity (EC)	EPA 120.1 and	1.60	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	51.1	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1020	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 17.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W3-1 - 2936 W. Monte Vista	T3346002F	Electrical Conductivity (EC)	EPA 120.1 and	1.35	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	8.09	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	865	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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SOURCE: Well Water

NOTES: Temperature above qualifying at 17.4 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W3-2 - 2848 W. Monte Vista	T3346002G	Electrical Conductivity (EC)	EPA 120.1 and	1.93	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	35.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1240	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.090
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 19.8 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W4-1 - 6618 Jennings	T3346002H	Electrical Conductivity (EC)	EPA 120.1 and	1.56	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	30.7	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	998	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 20.2 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W4-2 - 6800 Jennings	T3346002I	Electrical Conductivity (EC)	EPA 120.1 and	1.32	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	19.2	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	842	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/12/23 11:54 AM
SUBMITTED BY: Mike
ANALYZED DATE: 12/12/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature above qualifying at 17.6 C

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W5-1 - 6619 Jennings	T3346002J	Electrical Conductivity (EC)	EPA 120.1 and	2.08	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	43.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	1330	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.2	101 (T3346002A)	99.4 (T3346002A)	1.20
Electrical Conductivity (EC)	99.1			0.703
Nitrate Nitrogen (NO3-N)	101	101 (T3346002D)	102 (T3346002D)	1.51

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ELAP Certificate No.: 2714

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Authorized Copy Released To: _____

Client: ANDREW Zylstra DAIRY INC
Mailing address: 2655 W Monte Vista Ave
(not facility address)
Phone#:

Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	Analysis: H1	H3	M3	F18	OTHER
1	12-12	10:53	W1-1 - DAIRY 1 - MILK BARN		17.8		H1				
2			W1-3 - 7800 Vivian		18.4						
3			W2-1 - DAIRY 2 - A		18.4						
4			W2-2 - DAIRY 2 - B		18.4						
5			W2-4 - 7855 S. CARPENTER		18.4						
6			W3-1 - 2936 W Monte Vista		18.2						
7			W3-2 - 2948 W Monte Vista		17.4						
8			W4-1 - 6618 Jennings		19.8						
9			W4-2 - 6866 Jennings		20.2						
10			W5-1 6619 Jennings		17.6						

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[signature]	Andrew Zylstra Dairy				
2nd		DENELE ANALYTICAL, INC.	12-12-23 @ 11:54			T 3346002

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/19/23 11:57 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.4c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W10-1 3831 N Dakota DW	T3353008A	Electrical Conductivity (EC)	EPA 120.1 and	0.837	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	12.8	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	536	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.2	99.4 (T3352006A)	98.2 (T3352006A)	1.21
Electrical Conductivity (EC)	98.9			0.750
Nitrate Nitrogen (NO3-N)	105	106 (T3353008A)	100 (T3353008A)	5.20

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ELAP Certificate No.: 2714

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/19/23 11:57 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 14.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W10-2 3831 N Dakota DW	T3353008B	Electrical Conductivity (EC)	EPA 120.1 and	0.961	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	12.5	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	615	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.2	99.4 (T3352006A)	98.2 (T3352006A)	1.21
Electrical Conductivity (EC)	98.9			0.750
Nitrate Nitrogen (NO3-N)	105	106 (T3353008A)	100 (T3353008A)	5.20

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/19/23 11:57 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/19/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 15.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W10-5 4154 Murphy DW	T3353008C	Electrical Conductivity (EC)	EPA 120.1 and	0.574	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	10.0	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	367	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	94.2	99.4 (T3352006A)	98.2 (T3352006A)	1.21
Electrical Conductivity (EC)	98.9			0.750
Nitrate Nitrogen (NO3-N)	105	106 (T3353008A)	100 (T3353008A)	5.20

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/19/23 11:57 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 14.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W10-4I Dakota IW	T3353008D	Electrical Conductivity (EC)	EPA 120.1 and	0.918	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	18.3	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	555	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	21.1	mg/L
		Soluble Salts (SALT-SOL)	Calculation	587	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	94.2	99.4 (T3352006A)	98.2 (T3352006A)			1.21
Electrical Conductivity (EC)	98.9			1.07 (T3352006A)	1.06 (Dup)	0.750
Nitrate Nitrogen (NO3-N)	105	106 (T3353008A)	100 (T3353008A)			5.20
Total Dissolved Solids (TDS)	98.5			615 (T3353011A)	620 (Dup)	0.810
Total Nitrogen (TN)	99.7			11.2 (T3338006E)	11.2 (null)	0.000

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Denele Analytical, Inc.

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Compliance Analysis Report

Customer
Andrew Zylstra Dairy
2655 W. Monte Vista Ave.
Modesto, CA 95358

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER:
RECEIVED DATE: 12/19/23 11:57 AM
SUBMITTED BY: Andrew
ANALYZED DATE: 12/19/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 15.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
W9-1I McCarthy IW	T3353008E	Electrical Conductivity (EC)	EPA 120.1 and	1.07	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	23.1	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	545	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	25.9	mg/L
		Soluble Salts (SALT-SOL)	Calculation	687	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	94.2	99.4 (T3352006A)	98.2 (T3352006A)			1.21
Electrical Conductivity (EC)	98.9			1.07 (T3352006A)	1.06 (Dup)	0.750
Nitrate Nitrogen (NO3-N)	105	106 (T3353008A)	100 (T3353008A)			5.20
Total Dissolved Solids (TDS)	98.5			615 (T3353011A)	620 (Dup)	0.810
Total Nitrogen (TN)	99.7			11.2 (T3338006E)	11.2 (null)	0.000

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DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: ANDREW ZYLSTRA DAIRY INC
Mailing address: 2655 W. Monte Vista AVE
(not facility address) Modesto CA 95356
Phone#:

Authorized Copy Released To: Copy F+R
Sampled By:
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

TILE DRAIN
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

DISCHARGE WATER
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

PLANT TISSUE
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	12/19/23	8:00 AM	W10-1 3831 N DAKOTA	DW	14.4°C		H1				
2	↓	↓	W10-2 3831 N DAKOTA		14.6°		↓				
3			W10-5 4154 Murphy		15.0°C						
4			W10-4E DAKOTA IRRIGATION WELL	IW		14.6°C					H7
5			W9-1E McCARTHY IRRIGATION WELL	IW		15.0°C					H7
6											
7											
8											
9											
10											

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged in By:	LABORATORY NUMBER
1st	[signature]					
2nd	[signature]	DENELE ANALYTICAL, INC.	12-19-23 10:57			T3353008

Required to be submitted to any agency? (YES / NO)

Zylstra Dairies 1 2023 Annual Report

Final Audit Report 2024-06-26

Created:	2024-06-26
By:	Joe Ramos (jramos@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAKT3rgYfM58SW4Qcv-SXYp5R3jhJqZ_Rd

"Zylstra Dairies 1 2023 Annual Report" History

Document created by Joe Ramos (jramos@fragservices.com)
2024-06-26 - 6:29:35 PM GMT

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Email viewed by zylstradairies@gmail.com
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Signer zylstradairies@gmail.com entered name at signing as mike zylstra
2024-06-26 - 7:22:32 PM GMT

Document e-signed by mike zylstra (zylstradairies@gmail.com)
Signature Date: 2024-06-26 - 7:22:34 PM GMT - Time Source: server

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